Image Deblurring using ILSTAT-ADMM

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Image Deblurring Problem

In imaging systems, the observed image is modeled as:

$$b = Ax + n$$

Where:

x: original image
A: Blur operator
b: observed blurred image
n: noise

Goal:

Recover x from b

Problem:

Blur removes high-frequency details, making inversion difficult.

Difference Between Blur and Noise

Blur

Blur occurswhen the imaging system spreads pixel intensity across neighboring pixels.

Mathematically:

$$b = Ax$$

Where

- A= blur operator
- x = original image

Blur is typically modeled as convolution with a kernel:

Example blur kernels:

- Gaussian blur
- Motion blur
- Defocus blur

Effect of Blur

- Loss of sharp edges
- Loss of high-frequency details
- Smooth appearance

Noise

Noiseisrandom disturbance added to the image signal.

Mathematical model:

$$b = Ax + n$$

Common noise types:

- Gaussian noise
- Salt & pepper noise
- Poisson noise

Effect of Noise

- Random pixel fluctuations
- Grainy appearance

Property	Blur	Noise
Cause	optical system	random disturbance
Mathematical model	convolution	additive term
Effect	smooth edges	random intensity

Blur Operator

Blur is represented as convolution:

$$Ax = h \times x$$

Where h is the Gaussian blur kernel.

Meaning:

- The operator A applies Gaussian blur to the image.

Compressed Sensing

Compressed sensingisbasedonthe idea that natural signals are sparse.
A signal x can be reconstructed from fewer measurements if it is sparse.

Mathematical formulation:

$$\min_{x} \|x\|_0$$

subject to

$$Ax = b$$

Where

- x = sparse signal
- A = measurement matrix
- b = observed measurements

Why Compressed Sensing Works

Natural images are sparse in transform domains:

- Wavelet
- DCT
- Gradient domain

Therefore only a small number of coefficients contain most information.

Connection to Our Project

Our optimization problem:

$$\min_{x} \|Ax - b\|^2 + \lambda \|x\|_0$$

This is a compressed sensing formulation that recovers sparse image representations.

Optimization Formulation

We solve the reconstruction problem by minimizing:

$$\min_{x} \|Ax - b\|^2 + \lambda \|x\|_0$$

Data fidelity

$$\|Ax - b\|^2$$

Ensures reconstructed image matches observation.

Sparsity regularization

$$\lambda \|x\|_0$$

Encourages sparse representation.

λ controls the **strength of regularization**.

Why Variable Splitting

Direct optimizationisdifficultbecause:

- Ax couples variables
- L0 norm is non-convex

Introduce auxiliary variable

$$y = Ax$$

Now the problem becomes

$$\min_{x,y} \|y - b\|^2 + \lambda \|x\|_0$$

subject to

$$y = Ax$$

This separates the two terms.

Augmented Lagrangian

We form the augmented Lagrangian:

$$L(x, y,\ z) = 1/2\|y - b\|^2 - z^t\,(y - Ax) + \frac{\rho}{2}\|y - Ax\|^2$$

Where

- z → Lagrange multiplier
- ρ → penalty parameter

Terms	Meaning
$\|y - b\|^2$	data fidelity
$z^t\,(y - Ax)$	constraint enforcement
$\rho\|y - Ax\|^2$	penalty term

x-Update Derivation

Minimize

$$\min_{x} \lambda \|x\|_0 + \frac{\rho}{2}\|y - Ax\|^2 - z^t (y - Ax)$$

Taking gradient:

$$\bigtriangledown f = A^T (Ax - y + z/\rho)$$

Gradient descent update:

$$x_{k+1} = x_k - \frac{1}{v} \bigtriangledown f$$

Then apply shrinkage operator:

$$x = sign(x) max(|x| - \lambda, 0)$$

This enforces sparsity.

y-Update Derivation

Minimize

$$\min_{y} \|y - b\|^2 + \frac{\rho}{2}\|y - Ax\|^2 - z^T (y - Ax)$$

Set derivative to zero:

$$\frac{\partial L}{\partial y} = 0$$

Solving gives

$$y = \frac{b + \rho Ax - z}{1 + \rho}$$

z-Update Derivation

Multiplier update:

$$z_{k+1} = z_k + \rho\left(y - Ax\right)$$

Purpose:

Enforces constraint

$$y \approx Ax$$

Complete Algorithm

At each iteration:

x-update

$$x_{k+1} = Shrink\left(x - \frac{1}{v} \bigtriangledown f\right)$$

y-update

$$y = \frac{b + \rho Ax - z}{1 + \rho}$$

z-update

$$z = z + \rho\left(y - Ax\right)$$

Repeat until convergence.

Limited Shrinkage Thresholding (LST)

LSTisathresholdingoperator used toenforce sparsity.
It removes small coefficients while preserving large ones.

General shrinkage operator:

$$Shrink\left(x,\lambda\right)=sign\left(x\right)max\left(\left|x\right|-\lambda,0\right)$$

In our algorithm this step promotes sparse solutions.

Role of LST in the Algorithm

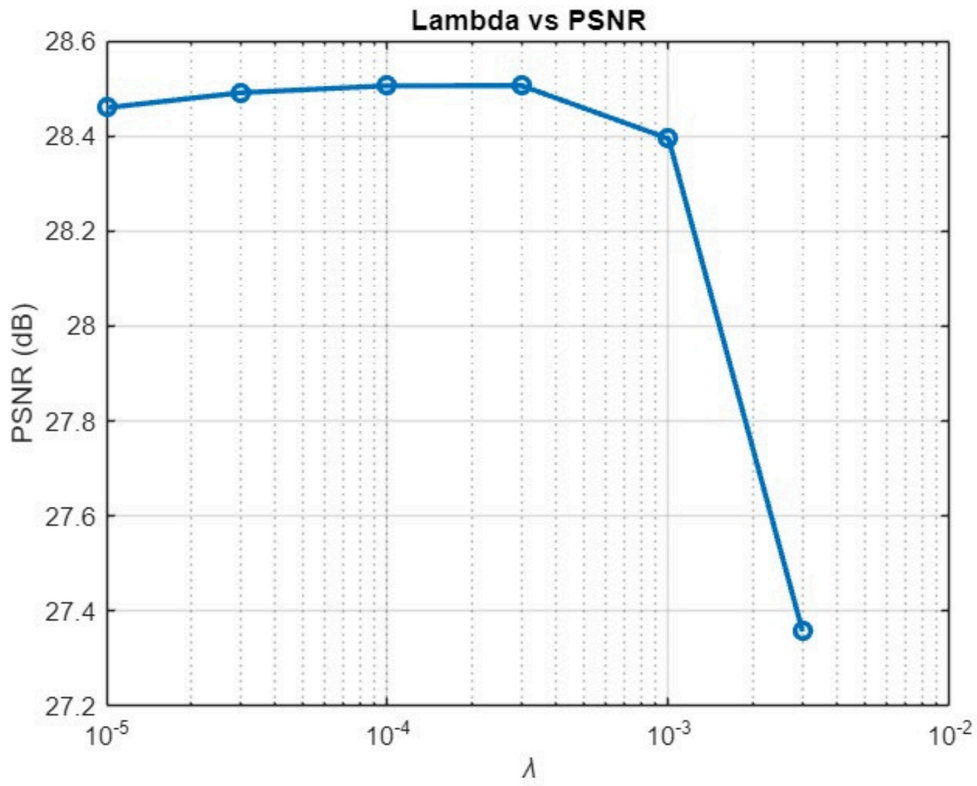
$$x=Shrink\left(x-\frac{1}{v}\bigtriangledown f\right)$$

Purpose:

- Remove noise
- Preserve important features
- Enforce sparsity

λ vs PSNR

Obser vation:



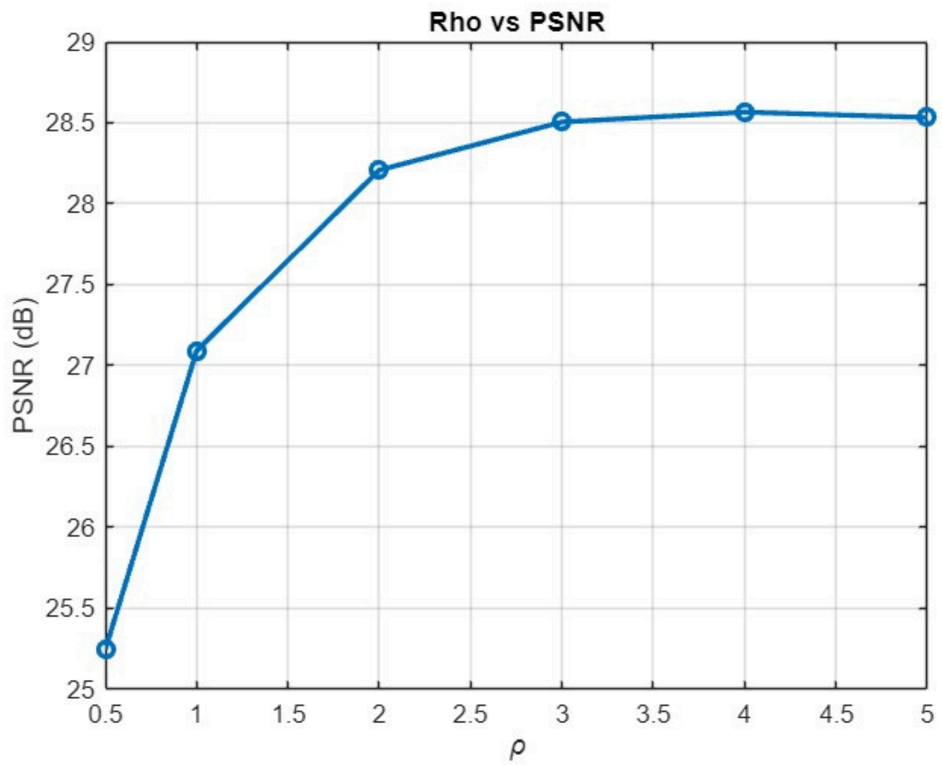
PSNR peaks near

$$\lambda = 3 \times 10^{-4}$$

Explanation:

- Small λ → preserves details
- Large λ → oversmoothing

ρ vs PSNR



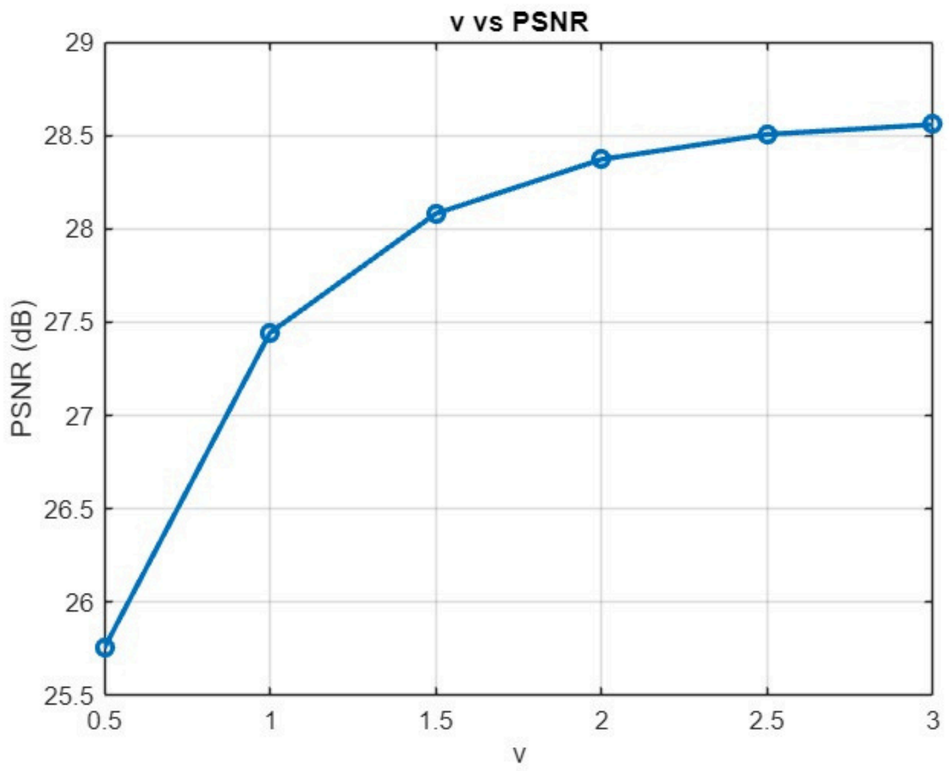
PSNR improves as ρ increases.

Optimal value:

$$\rho = 4$$

Large ρ enforces the constraint more strongly.

v vs PSNR



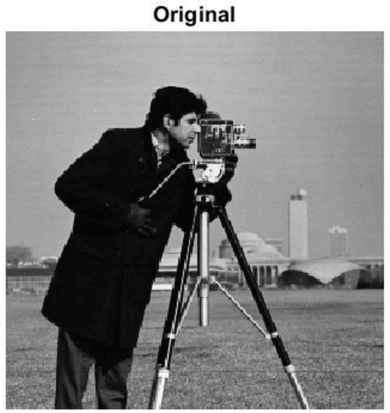
PSNR increases with v.

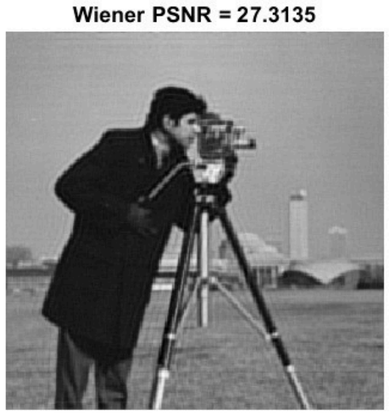
Optimal value:

$$v = 3$$

Controls gradient step size.

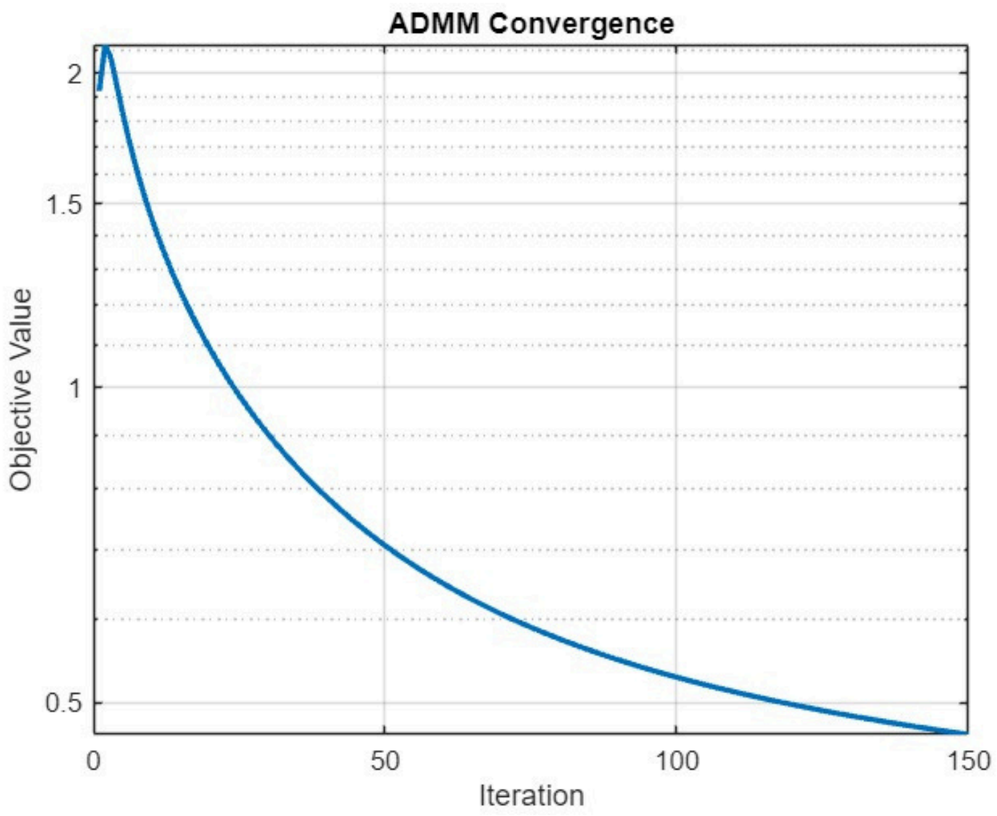
Comparison with Wiener Filter





ILSTAT improves reconstruction accuracy.

Convergence Analysis



- The objective function decreases with iterations.
- This shows the algorithm converges.

Peak Signal-to-Noise Ratio (PSNR)

PSNRmeasuresthe qualityof reconstructedimages.

First compute Mean Squared Error (MSE):

$$MSE = \frac{1}{N}\Sigma(x - x_{rec})^2$$

Then compute PSNR:

$$PSNR = 10log_{10}\left(\frac{MAX^2}{MSE}\right)$$

Where

- MAX = maximum pixel value (1 for normalized images)

Conclusion

ILSTAT-ADMMsuccessfully restores blurred images.

Key contributions:

- mathematical formulation
- ADMM optimization
- parameter tuning
- experimental validation

Final PSNR achieved:

28.53 dB